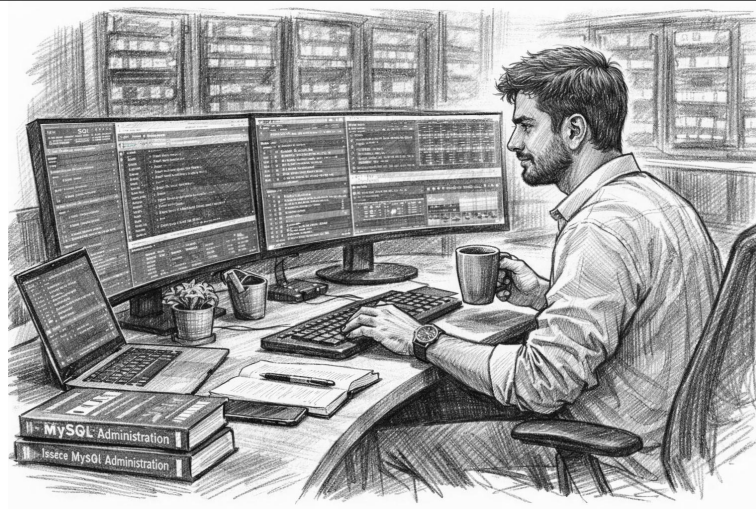




MySQL Day 26

MySQL Troubleshooting & Production Incident Handling

Website very slow

Database not responding

Replication stopped

Disk full

Too many connections

High CPU usage

Author : Ajit Yadav (MSSQL , MYSQL , POSTGRESQL DBA)

DBA Incident Response Flow

When u receive alert :

- 1 Confirm issue
- 2 Check MySQL status
- 3 Check system resources
- 4 Check running queries
- 5 Check logs
- 6 Identify root cause
- 7 Fix problem
- 8 Document incident

Soln:

Check mysql service .

First verify databas service is running ?

systemctl status mysql

```
root@db-01:/# systemctl status mysql.service
● mysql.service - MySQL Community Server
   Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-03-02 12:32:56 IST; 1 week 5 days ago
     Process: 1238862 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
    Main PID: 1238870 (mysqld)
      Status: "Server is operational"
        Tasks: 48 (limit: 9286)
       Memory: 3.6G
         CPU: 3h 37min 46.535s
      CGroup: /system.slice/mysql.service
              └─1238870 /usr/sbin/mysqld
```

if stop start is using : systemctl start mysql

Check Server Resource Usage

CPU Usage: using below command

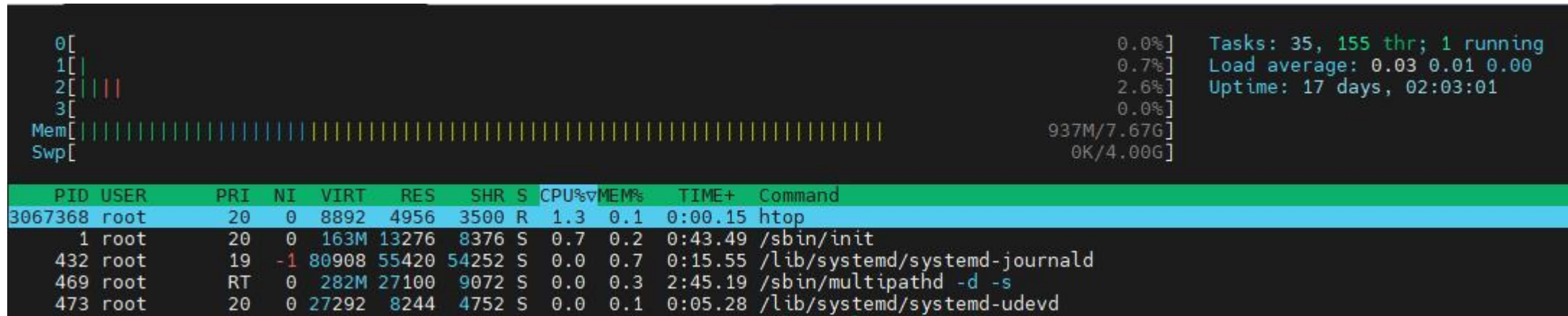
top & htop

```
root@db-01:/# top
top - 13:49:02 up 17 days, 2:02, 2 users, load average: 0.06, 0.02, 0.00
Tasks: 132 total, 1 running, 131 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.5 sy, 0.0 ni, 99.3 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7853.9 total, 2151.8 free, 935.7 used, 4766.4 buff/cache
MiB Swap: 4096.0 total, 4096.0 free, 0.0 used. 6616.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1238870	mysql	20	0	2894656	656988	40552	S	1.0	8.2	217:47.91	mysqld
1	root	20	0	167704	13276	8376	S	0.0	0.2	0:43.48	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.36	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp

htop

root@db-01:/# htop



Memory usage :

```
root@db-01:/# free -m
```

	total	used	free	shared	buff/cache	available
Mem:	7853	940	2145	1	4767	6610
Swap:	4095	0	4095			

root@db-01:/#

Disk usage

```
root@db-01:/# df -kh
```

Filesystem	Size	Used	Avail	Use%	Mounted on
tmpfs	786M	1.4M	785M	1%	/run
/dev/mapper/ubuntu--vg-ubuntu--lv	455G	14G	441G	4%	/
tmpfs	3.9G	0	3.9G	0%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock
/dev/sda2	2.0G	257M	1.6G	15%	/boot
/dev/sda1	1.1G	6.1M	1.1G	1%	/boot/efi
tmpfs	786M	4.0K	786M	1%	/run/user/0

root@db-01:/#

Check Running queries:

show processlist;

login to mysql:

```
mysql> show processlist;
```

Id	User	Host	db	Command	Time	State	Info
7	event_scheduler	localhost	NULL	Daemon	1041599	Waiting on empty queue	NULL
18	repl_user	db-02:37464	NULL	Binlog Dump GTID	1038139	Source has sent all binlog to replica; waiting for more updates	NULL
7214	root	localhost	NULL	Query	0	init	show processlist

```
3 rows in set, 1 warning (0.00 sec)
```

If query running very long → possible issue.

kill if necessary

kill 18

CREATE INDEX idx_email ON users(email);

and check again.

Identify Slow Queries

First enable slow query log & Check slow query log.

/var/log/mysql/slow.log

view log

tail -f /var/log/mysql/slow.log

ex: SELECT * FROM users WHERE email='test@gmail.com';

Check MySQL Error Log

tail -f /var/log/mysql/error.log

```
root@db-01:/# tail -f /var/log/mysql/error.log
```

Replication Troubleshooting

SHOW REPLICA STATUS\G

Field	Meaning
Replica_IO_Running	IO thread
Replica_SQL_Running	SQL thread
Seconds_Behind_Master	replication lag

Detect Deadlocks

Deadlocks happen when transactions block each other.

Check:

SHOW ENGINE INNODB STATUS;

```
mysql> SHOW ENGINE INNODB STATUS\G;
***** 1. row *****
Type: InnoDB
Name:
Status:
=====
2026-03-14 14:51:30 140105593439808 INNODB MONITOR OUTPUT
=====
Per second averages calculated from the last 11 seconds
-----
BACKGROUND THREAD
-----
srv_master_thread loops: 4060 srv_active, 0 srv_shutdown, 1040320 srv_idle
srv_master_thread log flush and writes: 0
-----
SEMAPHORES
-----
OS WAIT ARRAY INFO: reservation count 15534
OS WAIT ARRAY INFO: signal count 14542
RW-shared spins 0, rounds 0, OS waits 0
RW-excl spins 0, rounds 0, OS waits 0
RW-sx spins 0, rounds 0, OS waits 0
Spin rounds per wait: 0.00 RW-shared, 0.00 RW-excl, 0.00 RW-sx
-----
TRANSACTIONS
-----
Trx id counter 5262569
Purge done for trx's n:o < 5262566 undo n:o < 0 state: running but idle
History list length 0
LIST OF TRANSACTIONS FOR EACH SESSION:
---TRANSACTION 421581092355288, not started
0 lock struct(s), heap size 1128, 0 row lock(s)
---TRANSACTION 421581092354480, not started
0 lock struct(s), heap size 1128, 0 row lock(s)
---TRANSACTION 421581092353672, not started
0 lock struct(s), heap size 1128, 0 row lock(s)
```

Too Many Connections

Check connection timeout
SHOW STATUS LIKE 'Threads_connected';

Fix options:
Increase max_connections
Optimize connection pooling
Kill idle sessions

Disk Full Issue:
check disk space

df -kh
Binary logs growing
Large backup files
Slow query logs

```
root@db-01:~# df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            786M   1.4M   785M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 455G   14G   419G   4% /
tmpfs            3.9G     0   3.9G   0% /dev/shm
tmpfs            5.0M     0   5.0M   0% /run/lock
/dev/sda2        2.0G   257M   1.6G  15% /boot
/dev/sda1        1.1G    6.1M   1.1G   1% /boot/efi
tmpfs            786M   4.0K   786M   1% /run/user/0
root@db-01:~#
```

PURGE BINARY LOGS BEFORE NOW()-INTERVAL7DAY;

```
mysql> show binary logs;
+-----+-----+-----+
| Log_name | File_size | Encrypted |
+-----+-----+-----+
| mysql-bin.000040 | 237 | No |
| mysql-bin.000041 | 284 | No |
| mysql-bin.000042 | 284 | No |
| mysql-bin.000043 | 260 | No |
| mysql-bin.000044 | 864 | No |
| mysql-bin.000045 | 324 | No |
| mysql-bin.000046 | 324 | No |
| mysql-bin.000047 | 894 | No |
| mysql-bin.000048 | 1956 | No |
| mysql-bin.000049 | 324 | No |
| mysql-bin.000050 | 324 | No |
| mysql-bin.000051 | 105190762 | No |
| mysql-bin.000052 | 104994005 | No |
| mysql-bin.000053 | 105272070 | No |
| mysql-bin.000054 | 25157876 | No |
| mysql-bin.000055 | 3647 | No |
| mysql-bin.000056 | 109595647 | No |
| mysql-bin.000057 | 109603218 | No |
| mysql-bin.000058 | 109600399 | No |
| mysql-bin.000059 | 109604433 | No |
| mysql-bin.000060 | 109597360 | No |
| mysql-bin.000061 | 221294782 | No |
| mysql-bin.000062 | 216611063 | No |
| mysql-bin.000063 | 111505028 | No |
| mysql-bin.000064 | 111465850 | No |
| mysql-bin.000065 | 6216 | No |
| mysql-bin.000066 | 111467839 | No |
| mysql-bin.000067 | 111465671 | No |
| mysql-bin.000068 | 111485779 | No |
| mysql-bin.000069 | 27618 | No |
| mysql-bin.000070 | 105026716 | No |
| mysql-bin.000071 | 105002234 | No |
| mysql-bin.000072 | 104948776 | No |
| mysql-bin.000073 | 169601208 | No |
| mysql-bin.000074 | 3364 | No |
| mysql-bin.000075 | 277 | No |
+-----+-----+-----+
36 rows in set (0.00 sec)
```

```
mysql> purge binary logs before now()-interval 5 day;
Query OK, 0 rows affected (0.21 sec)
```